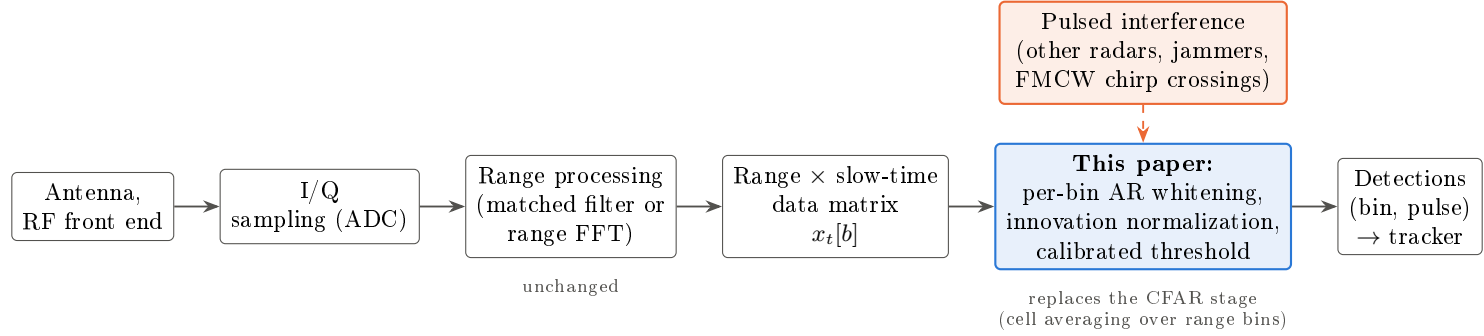


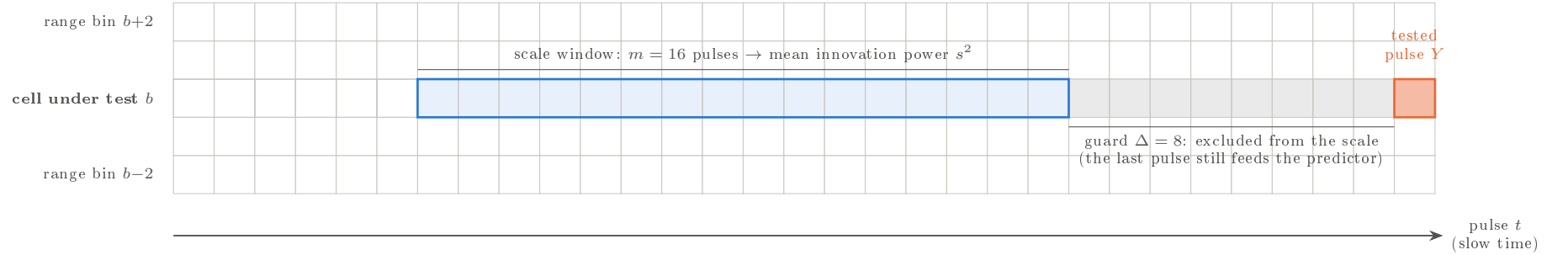
Innovation-normalized detection: what it does in the radar signal chain

Each range bin is processed on its own, along slow time (pulse to pulse). The threshold does not use neighbouring range bins.

(a) Where it fits in the receive chain

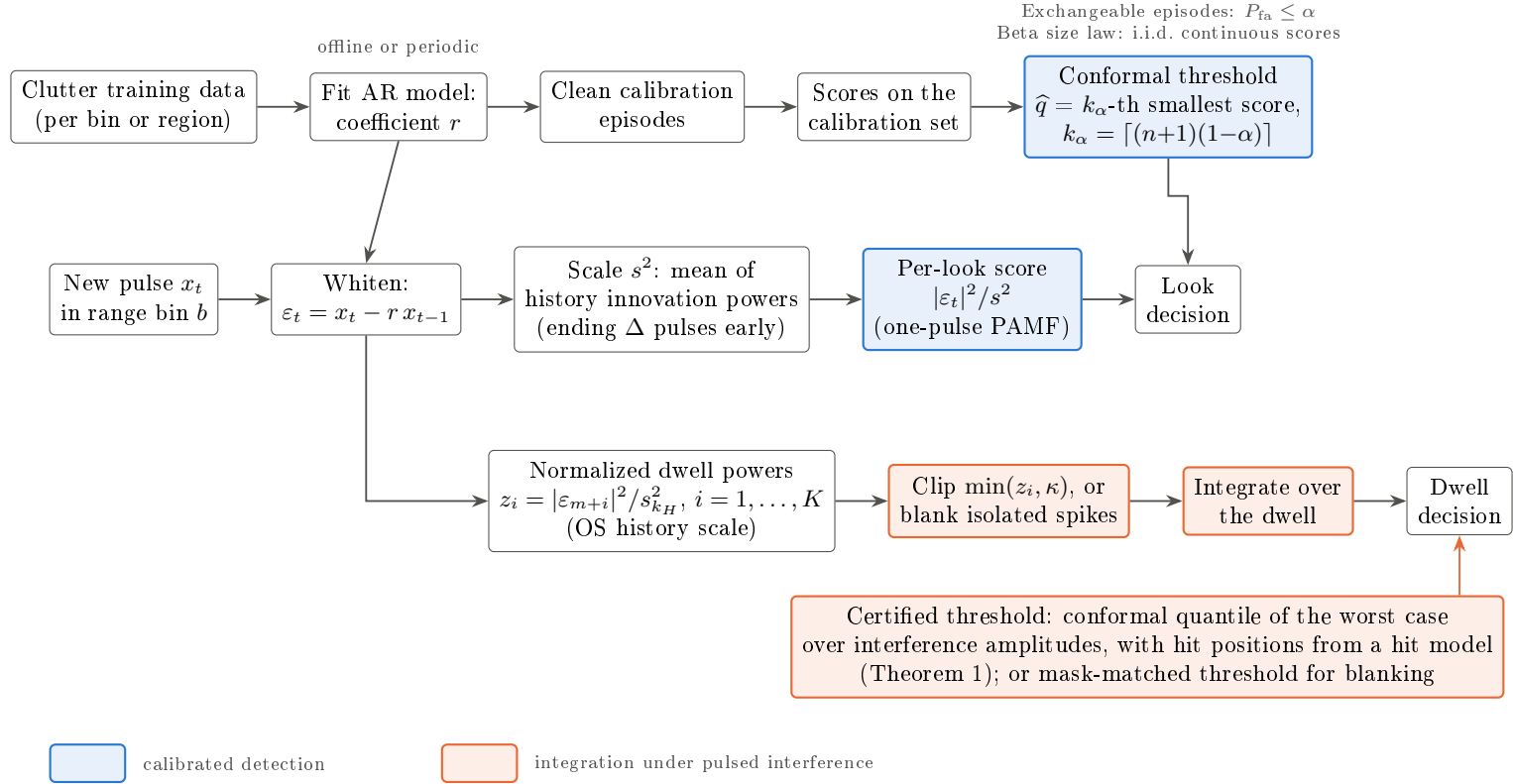


(b) What happens to one range bin: the slow-time window of one test



$\varepsilon_t = x_t - r x_{t-1}$ (AR(1) innovation; r fitted on clutter training data), $\text{score} = |\varepsilon_Y|^2 / s^2$, detection if $\text{score} > \hat{q}^2$ (conformal threshold).
 A persistent target enters the unguarded scale after onset and can mask itself. Under the model, a guard delays scale contamination by Δ looks.

(c) Inside the detector



Reading the diagram. Range CA-CFAR relies on representative reference cells. In correlated, textured clutter, each bin's own *past* provides an alternative reference. At the true AR model, with a common texture and no thermal noise, whitening and innovation normalization remove the local clutter scale. A split-conformal threshold needs no Gaussian tail model, but its false-alarm bound requires exchangeable calibration and test episodes. A slow-time guard delays target contamination of the scale while retaining the latest sample in the predictor. For dwell detection, clipped or binary integration can use a certified threshold; pulse blanking is a separate, uncertified alternative. Certification covers arbitrary amplitudes when clean episodes are exchangeable, the test hit pattern is independent of all clean episodes and inclusion-dominated by the calibration hit law, and calibration masks are i.i.d. and independent of all clean episodes and the test hit pattern. These statistical conditions determine the mathematical scope.